INTERNSHIP PROJECT REPORT

AI-Based Solar PV Power Generation Forecasting Under Variable Weather Conditions

Submitted by: _____

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ABSTRACT

Solar photovoltaic (PV) power output changes with solar irradiance, temperature, humidity, wind and time of day. Accurate forecasting helps operators plan energy use, battery charging and grid integration. This project develops a machine-learning workflow to estimate PV power from weather and time-related inputs. A documented simulated hourly dataset is used to demonstrate data cleaning, exploratory data analysis, feature engineering, model training and evaluation. Linear Regression and Random Forest Regression are compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). A Streamlit dashboard lets a user enter weather conditions and view an estimated output in watts and kilowatts. Because the dataset is simulated, the project demonstrates the method rather than proving real-world plant accuracy.

1. INTRODUCTION

Solar energy is a renewable energy source that converts sunlight into electrical energy using photovoltaic modules. PV output is not constant because sunlight and environmental conditions change throughout the day. Clouds can reduce irradiance quickly, while high panel temperature can lower conversion efficiency. Forecasting can support energy scheduling, storage management and monitoring. Machine learning can learn patterns from historical weather and power data and use them to estimate future output.

2. PROBLEM STATEMENT

Solar PV power generation varies under changing weather conditions. A simple fixed estimate cannot represent these changes reliably. The project therefore builds and compares regression models that use weather, time and recent output information to predict solar power for a defined time interval.

3. OBJECTIVES

1. Prepare a dataset containing PV power and weather measurements.
2. Clean missing values, duplicates and inconsistent timestamps.
3. Explore relationships between weather variables and PV output.
4. Create time-based and lag features.
5. Train at least two regression models.
6. Compare model performance with MAE, RMSE and R^2.
7. Develop a dashboard for interactive solar power prediction.

4. SCOPE AND DATASET

The included dataset contains hourly records for one year and represents an illustrative 5 kW PV system. Inputs include solar irradiance (W/m^2), ambient temperature (°C), panel temperature (°C), relative humidity (%), wind speed (m/s), hour, day of year, month and previous-hour power (W). The target is solar power output in watts. The data are simulated using a transparent educational formula with random variation; they are not measurements from a real plant. A real deployment should use validated site-specific historical PV and weather data.

5. SYSTEM ARCHITECTURE

Weather and historical PV data → Data cleaning → Exploratory data analysis → Feature engineering → Time-ordered train/test split → Train Linear Regression and Random Forest → Evaluate MAE, RMSE and R^2 → Save the best model → Streamlit dashboard → Predicted PV output.

6. METHODOLOGY

Data preprocessing: timestamps are parsed, records are sorted, duplicate timestamps are removed, and missing numerical inputs are handled with median imputation inside each model pipeline. Feature engineering: hour, day of year, month and previous-hour output are extracted. To reduce time-series leakage, the first 80% of chronologically ordered samples are used for training and the final 20% for testing. Linear Regression provides a simple baseline. Random Forest learns nonlinear relationships using an ensemble of decision trees. Predictions are clipped to the

illustrative system range of 0–5000 W.

7. EXPLORATORY DATA ANALYSIS

Recommended EDA plots include: (a) solar power over time, (b) irradiance versus power scatter plot, (c) average power by hour of day, and (d) correlation heatmap. Expected physical trends are that output is near zero at night and generally rises with irradiance during daylight. The strength of these trends depends on the simulated data generation assumptions and must be rechecked on real data.

8. MODEL EVALUATION

MAE is the average absolute difference between actual and predicted power. RMSE is the square root of the mean squared error and gives greater weight to large errors. R^2 measures how much target variation is explained relative to a mean baseline; values closer to 1 generally indicate better fit, while negative values can occur when a model performs worse than the baseline. The training script exports a model comparison table and actual-versus-predicted graph after execution. Numerical metrics should be taken from the generated `outputs/model_comparison.csv`, not guessed or reported before running the code.

9. DASHBOARD

The Streamlit application accepts irradiance, ambient temperature, panel temperature, humidity, wind speed, hour, day of year and previous-hour power. On clicking “Predict solar power,” the trained model displays the estimated power in watts and kilowatts. It also displays the model comparison table and the actual-versus-predicted plot when available. The dashboard is an educational demonstration and should not be used for operational decisions without validation.

10. RESULTS AND DISCUSSION

After running `python train_model.py`, compare the two models using the generated MAE, RMSE and R^2 values. The model with the lowest RMSE is selected as the best model in this implementation. Discuss whether the predicted curve follows the actual curve, whether errors are larger during rapid weather changes, and whether the model performs differently at low and high output. Since the data are simulated, the results measure performance on this synthetic dataset only and cannot be generalized to a real PV installation.

11. LIMITATIONS

The dataset is simulated and uses simplified assumptions. It does not fully represent shading, inverter losses, equipment degradation, sensor faults, local cloud dynamics or plant-specific behavior. Random splitting is avoided, but a single chronological holdout is not enough to establish long-term robustness. Real use requires verified data, additional seasonal validation, monitoring for data drift and safety checks before connecting forecasts to control systems.

12. FUTURE ENHANCEMENTS

Use real site-level SCADA and weather data; add numerical weather forecasts, cloud cover and solar zenith angle; test Gradient Boosting or XGBoost; use rolling-origin cross-validation; provide prediction intervals; monitor data drift; and deploy the dashboard on a secured server. Forecast horizons such as 15 minutes, 1 hour or 24 hours should be defined explicitly and evaluated separately.

13. CONCLUSION

This project demonstrates an end-to-end machine-learning workflow for solar PV power forecasting under variable weather conditions. It includes preprocessing, feature engineering, two regression models, standard evaluation metrics and an interactive dashboard. The simulated dataset makes the project reproducible for an internship demonstration. Real-world accuracy must be established by replacing the synthetic dataset with measured PV and weather records and evaluating the model on unseen periods.

REFERENCES

1. scikit-learn documentation, https://scikit-learn.org/

2. Pandas documentation, https://pandas.pydata.org/docs/

3. Streamlit documentation, https://docs.streamlit.io/

4. NREL, PVWatts Calculator and solar resource information, https://pvwatts.nrel.gov/ (for further real-data context).

APPENDIX: HOW TO RUN

1. Install Python 3.10 or later.

2. Run: pip install -r requirements.txt

3. Run: python train_model.py

4. Run: streamlit run app.py

5. Review outputs/model_comparison.csv and generated plots.